

The Relational Context Layer: A Reference Architecture for AI Support of Response-Mediated Work

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Abstract

Large language models are stateless with respect to the user's relational world: each interaction begins without a persistent model of the people the user must work with and through, of the user's own goals and constraints, or of the structural conditions that bound what the user can achieve. This caps the quality of relational assistance at that obtainable from a capable interlocutor who knows nothing of the user's situation, and it is an architectural property, not a deficit that scale will close. The setting is support of response-mediated work: professional tasks whose success is mediated by other people's responses, whether or not the work is interpersonal in form.

I propose the Relational Context Layer (RCL), a reference architecture for systems supporting such work — architecture in the systems-engineering sense of components, layers, and a maintenance regime arranged to meet constraints prior arrangements did not, not in the formal sense of a new representational grammar. RCL integrates a first-person model of the served human (navigator layer), other actors held as context-conditioned estimates of what they want (relational layer), and the structural conditions bounding the possibility space (contextual layer), maintained over the accumulated record of the relationship.

Formally, RCL is expressible as a finitely nested interactive POMDP under an epistemic reward, with the served human as one modelled agent and a dimension-specifically discounted self-report channel. I state this mapping rather than contest it. The formalism has existed for two decades; it was never deployed at realistic scale because exact solution is intractable and because no machinery existed to populate its distributions from the unstructured social evidence of real work — email, meetings, tone, history. Large language models remove the population barrier, which is why near-neighbour systems are now appearing: the field is arriving not at the idea but at its feasibility. RCL is feasible where I-POMDP deployment was not because it asks for radically less — no policy optimisation (the output is a prediction delivered to a human, not an action selected from a planning tree), recursion bounded at two to three levels on behavioural grounds, and approximate inference in place of exact belief update.

The architecture's central empirical bet, which no reduction can absorb because it concerns the world the model faces rather than any representation of it, is longitudinal: much of what makes an actor's behaviour interpretable is history-constituted, carried by the relationship's record and absent from the present moment, so a system that severs want-inference from that record predicts worse — most sharply on exactly the actions whose meaning lives in the history. The architecture's signature parameterisation, motivated by established findings about the humans the system models, is to distrust the served human's self-account on evaluative, ego-involved dimensions while trusting it on internal wants — a modelling choice no integrated-agent practitioner makes by default, and one settled by ablation rather than argument.

Claims carry explicit maturity tags; the relational layer is operationalised to the point of falsifiability; the validation programme specifies corpus, metrics, effect-size thresholds, and decision rules in advance. An early prototype, ReactAI, in which the author has a commercial interest, is noted as provenance only.

Keywords: relational modelling; AI support systems; agent memory; machine theory of mind; latent motivational state; longitudinal inference; interactive POMDP; ablation.

1. Introduction

A professional handling routine work does not treat each task as self-contained. A reply to a supplier querying an unpaid invoice is not written the same way to every supplier: the response is shaped by whether this is a relationship the firm can afford to lose, by whether the recent tone has been cooling, by what the writer knows the recipient actually cares about. None of that is in the email. It is supplied by a working model of the other party — what they want, how they relate to the writer, what they are likely to do next — that the professional maintains and revises continuously, largely without noticing. The same model governs more consequential cases: which of several stakeholders to approach, in what way, to advance an objective whose outcome depends on people who may not yet be involved.

Cases like these differ enormously in stakes, but they share a structure that is the subject of this paper: each is work whose success is mediated by another person's response. The supplier email succeeds or fails by how the supplier reacts to it; the stakeholder approach by how the stakeholder receives it. This holds whether or not the work looks interpersonal on its face — a financial analysis written to persuade a sceptical board, a report framed to survive a particular reader, are response-mediated in exactly the same sense as a negotiation, even though their surface form is a document rather than a conversation. I call this class *response-mediated work*. It is broad: it excludes only work whose quality is settled without reference to any person's reaction — code that only its author will run, a calculation checked against a known answer. Within the class, the everyday case and the high-stakes case are alike in kind and differ only in stakes and in frequency; the everyday case recurs orders of magnitude more often, which is where most of the aggregate value lies. Even tasks that present as purely quantitative mostly sit inside the class rather than outside it: a cash-flow projection has fixed arithmetic, but which projection one builds — which assumptions to adopt, which sensitivities to foreground, how to frame the result — is governed by the anticipated reaction of the people the projection is for. The response-mediation lives not in the calculation but in the choices that surround it, and those choices are most of the work.

A large language model brings almost none of the governing model to these situations. Each conversation begins without a representation of the user's relational world: no record of the supplier's cooling tone, no model of which counterparty holds real authority, no history between the user and the people they must work through. The user can supply context within a session, and the system will use it competently; but it does not persist, and the next interaction begins blank. This is not a limitation that scale or fine-tuning will remove. It is structural — contemporary systems are stateless with respect to the user's relational world — and because the quality of relational assistance depends almost entirely on how well the adviser models the people involved, a stateless system is capped, however capable the underlying model, at the quality of assistance obtainable from a capable interlocutor who knows nothing of the user's situation.

This carries a consequence for how widely such systems are used, which I state as a numbered claim because it is the most consequential and most testable bet the paper makes, and I want it exposed as such rather than buried in prose.

Claim 1 (the delegation conjecture; working position, falsifiable). If most professional work is response-mediated, and a stateless system must have the relevant relational context reconstructed by hand at the start of every interaction, then the binding constraint on how much of that work an AI system is actually used for is not the capability of the underlying model but the absence of a

maintained relational context and the cost of supplying it afresh each time. A capable user delegates only a fraction of the work they could — not because the model cannot do it, but because the cost of telling the model what it would need to know about the people the work is for exceeds the benefit, every time.

The claim is falsifiable: hold model capability fixed, vary whether a maintained relational context is available, and measure whether the share of real work a user will delegate — or the quality of the output they would actually submit — rises with it. If neither increases when the relational context is supplied and maintained, the claim is wrong, and the limit on AI's reach in professional work lies elsewhere.

A reference architecture, in the engineering sense

This paper characterises the representation a supporting system must maintain to lift the stateless cap, and proposes the *Relational Context Layer* (RCL) as that representation. I use the word *architecture* deliberately and in a specific sense, because the alternative sense would overclaim and the paper's later positioning depends on the distinction. By a reference architecture I mean an arrangement, in the systems-engineering sense, of components, layers, data flow, and a maintenance regime, specified so as to extend what a class of systems can do — not, in the formal sense, a new representational grammar or a primitive that prior formalisms could not express. The distinction is not evasive; it is the honest description of the contribution, and it has good precedent. The Transformer is reducible to compositions of existing operations, and MemGPT to a paging discipline over a context window; both are reducible to prior machinery and both are unproblematically architecture contributions, because the contribution lives in the arrangement meeting constraints that prior arrangements did not. RCL is an architecture contribution in exactly that sense and no stronger one.

I am explicit about the reducibility from the outset rather than defending against it later. RCL is expressible as a finitely nested interactive POMDP (Gmytrasiewicz & Doshi, 2005) under an epistemic reward, with the served human as one modelled agent among the others and a dimension-specifically discounted self-report channel; the detailed mapping is given in §2.3. Stating this concedes nothing the contribution needs. Expressibility within a near-universal formalism for rational-agent problems is a weak property — every algorithm is "just" an instantiation of a Turing machine — and it disposes of the formal sense of *architecture* without touching the engineering sense, which is the sense claimed.

Why now

Owning the reduction immediately raises the question the paper must answer in its own voice: if the formalism that integrates self, others, environment, and recursion has existed for two decades, why has none of the systems surveyed in §2 instantiated it for work of this kind? The answer is not that the idea was overlooked. It is that until recently the formalism was, at realistic scale, both unsolvable and unpopulatable — and the second of those barriers has only just fallen. Three reasons, in descending order of weight, together give the gap a clean explanation.

The first is computational intractability. Exact solution of a POMDP is PSPACE-hard, and the interactive POMDP compounds this with an interactive state space that grows explosively with the number of agents and the depth of nesting. Two decades of I-POMDP research live almost entirely in toy domains — multi-agent tiger problems, small gridworlds. The formalism was never neglected; it was never tractable at the scale real work demands.

The second, and the deep one, is the specification problem. Even granted unlimited compute, an I-POMDP must be authored: its states, dynamics, agent types, rewards, and observation models have to be written down. For a real workplace — where the evidence about actors arrives as unstructured

email, meetings, tone, and accumulated history — no machinery existed to convert that natural-language stream of social evidence into the distributions the formalism consumes. The formalism sat expressively complete and practically unpopulatable: able in principle to represent the problem, unable in practice to be filled with a real one.

The third is a matter of disjoint lineages. Decision-theoretic planning was largely bypassed by model-free deep reinforcement learning; the agent-memory systems surveyed in §2.1 descend from a third tradition entirely — databases and information retrieval — with little evident contact with the decision-theoretic literature. One community owned the formalism and never built for real workloads; another built for real workloads and had no occasion to reach for the formalism.

What changed, circa 2022, is that large language models became the first plausible engine for precisely the conversion the specification problem demands: inferring latent states, types, and motivations from unstructured text at scale. The barrier RCL's setting ran into was not the absence of the idea but the absence of the component that makes the idea buildable, and that component has existed for roughly three years. This also re-explains the convergence visible across 2025–26 — the near-neighbour systems named in §2 — as the field arriving not at the idea but at its feasibility. Stated once, the arc is this: the formalism for integrated multi-agent modelling has existed since 2005; it went undeployed because it was intractable in exact form and unpopulatable from real evidence; large language models remove the population barrier; and RCL is the engineering specification for what becomes buildable as a result.

This invites an immediate objection, and answering it surfaces a claim that does real work later in the paper. Intractability did not vanish when language models arrived — exact I-POMDP solution remains PSPACE-hard — so why is RCL buildable where I-POMDP deployment was not? The answer is not that the system is cheaper because the human it serves does less downstream thinking; that would credit the saving to the user's habits rather than to anything architectural, and it would make the claim depend on assumptions about the end user that it should not. The answer is that RCL computes a different and smaller object than an I-POMDP solver does, and the difference is intrinsic to it.

Claim 2 (support as tractability concession; working position). Prediction is computationally cheaper than planning, and RCL's output is a prediction. An I-POMDP agent must solve for a policy: it searches forward over the branching tree of its own action, the world's response, its next action, and so on, because the object it produces is a committed course of action whose later steps depend on earlier ones. That forward search over possible futures is the dominant cost of solving the formalism. RCL produces no policy and commits to no action. It computes an estimate — given what is now believed about an actor, what will that actor likely do — which is, in the limit, one step of inference from the current belief rather than a search over a tree of futures. The policy-optimisation burden therefore does not arise. This saving is a property of RCL itself, not of whatever consumes its output: replace the human navigator with an autonomous agent and RCL's cost is unchanged, because the planning cost, if it reappears, reappears in that agent as a separate component, not in RCL.

Two qualifications keep the claim in proportion. First, prediction is cheaper than planning but not free: the cost of nested belief update does not vanish, it is bounded — capped by the recursion limit of two to three levels (§5.4), which is a separate commitment grounded in how deep the modelled actors actually reason, not a consequence of this claim. The honest statement is that the dominant cost is removed and the remainder is bounded, not that prediction is trivial. Second, the saving is a tractability story for prediction-support specifically: a deployment that placed an exhaustively planning agent downstream of RCL would reintroduce the planning cost in that agent, and the framework claims no saving there. The support framing is therefore not only the ethical and deployment stance that §7

treats; it is a statement about computational budget, and it is the reason the bypass of §2.3 — obtaining the formalism's outputs without constructing its model — is affordable to run.

The keystone

With the reduction owned and the historical gap explained, the architecture's substantive bet can be stated — the one claim no reduction can absorb, because it concerns the world the model faces rather than any representation of it. The architecture integrates three elements: the user's own goals, constraints, and resources (the *navigator layer*); models of the other actors, held as context-conditioned estimates of their latent motivational states (the *relational layer*); and the structural conditions that bound what is achievable independently of any actor's preferences (the *contextual layer*). It lets each condition the recursion, and it maintains the integrated model over the accumulated record of the relationship. The load-bearing claim is the maintenance condition, and I state it in graded form deliberately, because the impossibility phrasing of earlier formulations claimed more than the evidence can carry.

Claim 3 (the keystone; working position, tested by ablation). A system supporting response-mediated work predicts worse — most sharply on the actions whose meaning is history-constituted — if it severs want-inference from the accumulated record of the relationship, or omits one of the three layers, or fails to let each condition the recursion. Much of what makes an actor's behaviour interpretable has no signature in the present moment: a curt three-word reply is hostility from one colleague and unremarkable habit from another, and only the record of prior exchanges resolves which. A system reasoning from a perfectly current snapshot still misreads that reply, because what it means is not present in the snapshot. The claim is about accuracy, not possibility — people perform recursive social reasoning without explicit self- or context-models all the time, frequently badly — and it is tested by ablation: remove a layer, or truncate the record toward an episodic snapshot, and prediction should degrade against held-out outcomes (§6).

Integration and longitudinal maintenance are the structure of this single claim, not two claims; and within it, it is the conditioning of want-inference on the accumulated record — not the bare persistence of facts, which agent-memory systems already supply — that bears the weight.

Contributions

This paper makes the following contributions, each carrying a maturity tag.

- **The longitudinal keystone** (central contribution; working position; tested by ablation): that conditioning want-inference on the accumulated record of a relationship, within an integrated three-layer model, materially improves recursive prediction in response-mediated work — because much of what makes behaviour interpretable is history-constituted and absent from the present moment. This is the one empirical bet the I-POMDP reduction cannot absorb, and the part of the contribution with independent life.
- **The why-now account and the support-as-tractability concession** (working position): a historical argument locating the contribution as an engineering specification for what large language models newly make buildable — the formalism long existing but intractable and unpopulatable — together with the claim that prediction-for-a-human is computationally cheaper than planning-for-an-agent, which is part of why the support case is feasible where the autonomous case was not.
- **The navigator-discount parameterisation** (distinctive testable commitment; application and synthesis, not discovery): the served human's self-account is trusted on internal wants and distrusted on evaluative, ego-involved attributes, with that discount propagated through the

recursion. This is a parameter assignment no integrated-agent practitioner makes by default — motivated by established findings about the humans the system models (§4.5) and settled by ablation (§6.1), not asserted as a representational primitive.

- **A falsifiable operationalisation of the relational layer** (working position; constructed to be falsifiable): motivation as a persistent want-set reweighted by situation, conflict resolved at evaluation time rather than stored, a bounded pre-declared inventory of conditioning variables, and three functional forms by which those variables act — each specified so that evidence can count against it (§5).
- **The delegation conjecture** (Claim 1 above; working position, falsifiable): that the binding constraint on AI's reach into professional work is the absence of a maintained relational context rather than model capability.
- **A pre-registration-grade validation programme** (methods contribution): a staged falsification scheme specifying corpus, baselines, metrics, effect-size thresholds, multiplicity control, and decision rules in advance, with the navigator-discount ablation as the minimum empirical increment (§6).

The tripartition of the model into three layers, and the claims that the three are jointly exhaustive and mutually irreducible, are retained as an *organising hypothesis* rather than as the contribution: they partition the architecture usefully and are tested as part of the ablation programme (§6), but the paper does not rest on them.

Scope

The keystone is, by its logic, general: if accurate recursive prediction requires these three layers maintained over a record in response-mediated work, the requirement holds of personal and social navigation as much as professional, since these present the same structure of a navigator, other actors with wants, and bounding conditions. I nonetheless confine this paper's development and examples to the professional domain, for two reasons that are not theoretical. First, professional environments constrain the range of acceptable behaviour by design — through role, norm, and procedure — which makes the relational field more tractable to model than the far wider behavioural range of personal life. Second, and more consequentially, the professional setting carries existing governance — codes of conduct, professional norms, institutional oversight — that bounds what advancing one's aims is permitted to mean, whereas the personal domain has no comparable structure and the same capability shades readily into the manipulation of intimates. I treat the ethical weight of that difference in §7.5. The generalisation of the claim beyond the professional domain is noted; the restriction of its application is a deliberate choice.

2. Related Work

Three research programmes bear on equipping an AI system to support a user's relational work, and taken together they locate the gap precisely. Agent memory supplies persistence but stores relational information as items. Machine theory of mind supplies latent-motivation inference but episodically, for an isolated other agent, without a self-model or a structural field. Integrated agent architectures supply exactly the integration of self, others, environment, and recursion that this paper builds on — but in service of an autonomous agent's own action, not a human's, and never built for the open human domain. I treat each in turn, engage the adjacent literatures more briefly, and then state the gap, bounding the claim to what a non-systematic survey can support.

2.1 Agent memory systems

That a language model retains nothing across sessions has prompted substantial work on external memory for language agents. MemGPT (Packer et al., 2023) treats the context window as analogous to physical memory in an operating system, paging information between a limited context and external storage to sustain coherence across long interactions. GraphRAG (Edge et al., 2024) constructs an entity–relationship–community graph over a document corpus and detects hierarchical community structure to support global queries. Zep/Graphiti (Rasmussen et al., 2025) — the current state of the art in graph-based agent memory — maintains a bi-temporal knowledge graph that separates the time an event occurred from the time it was recorded, with incremental update, contradiction detection, and edge invalidation. Mem0 (Chhikara et al., 2025) extracts and consolidates salient information from ongoing dialogue and, in its graph variant Mem0g, represents entities and their relations as a directed labelled graph.

These systems differ in sophistication, and the more advanced of them do more than the crude characterisation "they store facts" allows: Zep maintains continuously updated state rather than a static store, and modern systems can timestamp entries, record provenance, and reconcile contradictions. The point of the contrast that follows is therefore not that these systems are primitive at what they do. It is that the unit of representation remains the stored item — a node, an edge, a recorded episode — and storage, however well organised, is suited to the purpose these systems were built for: retrieving what is known. It is less suited to representing what an actor wants, which is not an item about them but a continuously revised estimate that shifts with circumstance, conflicts with itself, and is frequently unstated. A system that files a disposition the way it files an address has mis-typed the thing relational work most depends on — and the mis-typing is not repaired by storing the disposition with a timestamp and a confidence score, because what is missing is not metadata on a stored value but the conditioning of an inferred value on the situation and the record.

Structurally, each of these systems represents at most one of the three layers this paper argues are jointly necessary. GraphRAG models a document corpus — a contextual layer in our terms, and only partially. The others model a partial relational layer: a graph of entities and connections, without a motivational ontology. None represents the user's own goals and constraints as a distinct element, and none represents the structural field that bounds what the user can achieve. The point is not that these systems are deficient at what they do; it is that what they do is not what relational work requires.

2.2 Machine theory of mind

A second body of work comes considerably closer, and an honest account must say so plainly. Theory of mind — the capacity to attribute mental states such as beliefs, desires, and intentions to others and to predict behaviour from them (Premack & Woodruff, 1978) — has an active computational programme, and it is from this programme that the relational layer's central inferential move is adopted rather than invented. Bayesian theory of mind (Baker, Saxe & Tenenbaum, 2009, 2011) frames the inference of an agent's desires and beliefs from observed behaviour as Bayesian inverse planning: given actions, invert a model of rational action to recover the goals that best explain them, conditioning that inference on the environment by construction. This is precisely the representational move the relational layer requires — actors held as inferred wants, not stored items — and I take it as established machinery rather than claim it. Rabinowitz et al. (2018) gave the move a neural-network form, learning from observation alone to infer the latent characteristics of previously unseen agents and predict their actions.

The capability has advanced quickly. On standardised theory-of-mind batteries, contemporary large language models now match or exceed average human performance on several measures — false-belief understanding, interpreting indirect requests, recognising irony — though not uniformly (Strachan et al., 2024), and on higher-order tasks requiring recursively nested mental states recent

work reports models reaching adult human performance (Street et al., 2024). The interpretation of these results is contested — models fail on trivial perturbations of classic tasks (Ullman, 2023), and whether benchmark success reflects mental-state representation or pattern-matching remains open — but the framework does not need that debate resolved. The weaker claim suffices: inferring what an actor wants in order to predict what they will do is an active and capable line of work the framework builds on.

What this programme has not been built for is the shape of problem a professional navigating an ongoing relationship faces, and the limitation is sharper than "it does not persist." Machine-theory-of-mind systems are overwhelmingly episodic: they reason about a single interaction, a benchmark item, or a short simulated exchange, and about an isolated other agent. The cost of that is not merely that yesterday's evidence is unavailable; it is that a large part of what makes an action interpretable at all has no signature in the episode being reasoned about. A curt one-line reply reads as hostility from one colleague and as unremarkable habit from another; nothing in the message discloses which, and only the accumulated record of prior exchanges resolves it. Meaning of this kind is history-constituted — carried by the record of what came before, not by the present action — so a system reasoning episodically, however well it reasons within the episode, systematically misreads exactly the actions whose sense lives in the history. This is the mechanism behind the longitudinal requirement (§4.6): persistence is not bookkeeping carried out alongside the inference but a precondition of inferring correctly. Machine theory of mind supplies the inference and omits the record on which a large fraction of the inference depends. It also, by treating the observer as a neutral vantage point, models neither the user's own motivational structure as an element of the problem nor the structural field that bounds what any party can do.

2.3 Integrated agent architectures and the role of the self

The integration this paper argues for — self, other actors, the structural field, and recursive belief in one representation — is not unoccupied. It is most completely occupied by a literature the two programmes above do not touch, and the honest positioning of this framework turns on stating that plainly and then locating the difference precisely. The interactive POMDP (I-POMDP; Gmytrasiewicz & Doshi, 2005) generalises partially observable Markov decision processes to multi-agent settings by folding models of other agents — including their nested beliefs about oneself and about each other — into the deciding agent's state space. An I-POMDP agent maintains beliefs over the physical environment and over the recursively nested models of others, and solves for its own optimal policy on that basis. Self, others, environment, recursion: all four are present and integrated, and on the bare question of whether these can be held in one representation, I-POMDP answers it. What it does not settle — and what this paper reserves — is whether conditioning that representation on the accumulated record is necessary for the domain: a question not about the formalism, which can carry such conditioning, but about the world the model faces, and the one thing the reduction below cannot reach (Claim 4; §4.6).

I state the reduction explicitly, because a reviewer who knows the formalism will construct it, and admitting it is both more honest and more useful than defending against it. RCL is expressible as a finitely nested I-POMDP under the following instantiation: the deciding agent i is the system itself; its reward is epistemic — the accuracy of a prediction it emits about a designated actor's behaviour, a reward over its own posterior rather than over any world-state it acts to bring about; the served human is one modelled agent j among the others in the interactive state; and the prior over that human-model's type, together with the observation channel carrying the human's self-reports, is downweighted on a designated subset of evaluative, ego-involved dimensions. Under this instantiation, every component of the framework resolves into a value assignment within an existing slot of the formalism. The

navigator held as a distrusted actor is the prior on agent j plus a low-reliability observation model on those dimensions. The support framing — output a prediction, not an action — is the epistemic reward. These are choices within the machinery, not departures from it, and the framework claims no formal novelty over the formalism: the formal sense of the word *architecture* is disclaimed (§1).

The self–other modelling tradition (Raileanu et al., 2018) requires a separate note, because it invites a differentiator that does not hold. SOM predicts others by taking the agent's own model as a proxy — assume the other reasons as you do, then adjust — and it is tempting to distinguish RCL by saying it does not anchor on the self in this way. But the proxy-anchor is an approximation heuristic for the intractable belief update, not a commitment of the I-POMDP architecture: proper inverse planning over a non-degenerate prior is available to the formalism, which does not require the self-as-proxy. "Do not anchor on the self" is therefore "do not use SOM's approximation," not "leave the architecture," and it is not a differentiator.

The adequacy critique

Conceding the reduction settles what RCL is not. It does not settle the more interesting question, which the reduction sharpens: if the formalism expresses all of this, why was it never built for work of this kind, and what is RCL contributing by building it? The answer is an adequacy critique of the formalism's reach — not a claim that I-POMDP is inconsistent or expressively incomplete, which it is not, but the claim, in the form behavioural economics made against expected-utility theory, that its expressive completeness is purchased at the cost of requiring something its target domain cannot supply.

The cost is this. I-POMDP's expressive universality is purchased by requiring a complete finite specification of the interactive state — the actors, their types, the state space, the transition dynamics — before the formalism can be solved or even populated. This requirement is universal: it is the price of admission, paid in every application. Where the target domain natively admits such a specification, or admits a stipulated closure faithful enough for the purpose, the application succeeds — and real engineering systems do apply POMDP and bounded multi-agent models to robot localisation, scheduling, market microstructure, and adversarial routing, not by escaping the requirement but by stipulating a finite closure of the space and treating the remainder as noise. That the workaround is universally present is itself the evidence that the general form is inert: one does not apply I-POMDP to a real system, one applies it to a stipulated finite surrogate and relies on the surrogate being faithful.

Claim 4 (the no-faithful-closure position; working position; co-stands with the keystone). Open human domains admit no faithful finite closure. Any finite specification of the actors and what they want discards the history-constituted, continuously-revised content that drives the behaviour to be predicted — so the closure that is benign for a robot's floor plan is, for a counterparty, exactly the information that matters. The limitation is not that these domains are real and the formalism abstract; bounded mechanical and economic systems are real too and yield to stipulated closure. It is that open human domains are where the required closure is least available and most lossy. This claim is not asserted as self-evident, and it is not independent of the framework's central bet: if behaviour is history-constituted, with meaning carried by the accumulated record and absent from the present moment (the keystone, §1, §4.6), then no finite present-state closure can capture it, and Claim 4 follows. The two co-stand — exhibit a finite closure that predicts open human behaviour adequately, and both the adequacy critique and the keystone are damaged together.

This is what converts the reduction from the framework's vulnerability into its premise. RCL is expressible as an I-POMDP instantiation — formally, it is one — but it is implementable precisely because it never instantiates one: it does not enumerate the states, types, and dynamics the formalism requires and the open human domain cannot supply; it approximates predictions about actors directly from unstructured evidence, via a large language model, without constructing the complete interactive

model in between. Expressibility and implementability come apart here, and the gap between them is the contribution. The formalism could always express the problem; it could never be populated for an open human domain, because the population requires a closure that does not exist. What large language models provide is not a way to build the closure but a way to bypass it — to obtain the formalism's outputs, calibrated predictions about actors, without paying its specification cost, which no one could pay. The price of the bypass is the formalism's guarantees: with no constructed model there is no optimality proof, and the output is a defeasible prediction rather than a guaranteed-optimal policy. The support framing (§1) is what makes that price acceptable, and is also part of why the computation is feasible at all (Claim 2).

The remaining near-neighbours

Two recent systems sit close enough to require direct engagement. MetaMind (Zhang et al., 2025; NeurIPS 2025) is a staged social-reasoning system — theory-of-mind inference, then a moral/domain-norm constraint, then response generation — whose ablation shows each stage necessary. The overlap is the argument form: an architecture of several components, with ablation establishing each is required. I meet it on three points. First, the argument form is no one's property; establishing necessity by ablation is method, not contribution. Second, what is ablated differs in kind: MetaMind's three are processing stages a system passes through to produce one response in one interaction, whereas RCL's three are representational layers persistently co-resident and mutually conditioning, maintained over a record. Third, the middle terms are different categories — MetaMind's is a normative constraint (what response is morally or culturally appropriate), RCL's contextual layer a structural possibility-space constraint (what is achievable at all, regardless of preference). This is a category difference, which is the standard the positioning must meet. What separates RCL is not longitudinal maintenance — MetaMind's social memory accumulates the interlocutor's inferred beliefs, desires and emotional patterns across turns and conditions response generation on them — but structure: MetaMind models a single interlocutor, with no served navigator distinct from the parties that navigator must influence, no bias-discounted channel for the navigator's own self-report, and no recursion across those parties. The analogical first-to-third-person reasoning of the meta-ToM world model of Wang et al. (2026) — a distinct system that, confusingly, also carries the name *MetaMind* — inherits the self-as-proxy move addressed above and is differentiated on the same ground.

What distinguishes RCL in content, and how it is settled

After the reduction, one content difference remains that no integrated-agent practitioner makes by default, and it is the framework's signature modelling commitment rather than a representational primitive: the served human's self-account is distrusted on evaluative, ego-involved dimensions while trusted on internal wants, with that discount propagated through the recursion (§4.5). The status of this is precise. It is a parameter assignment — a skewed prior and a discounted observation channel — and the reduction shows it expressible. What makes it distinctive is not that it cannot be written down in the formalism but that no one writing down an I-POMDP for this setting would write it down by default: the default treats the principal's self-report as a reliable channel. The commitment is motivated by established findings about the humans the system models (§4.5), and it is settled — not by those findings and not by argument, but by the navigator-discount ablation of §6.1: build the model treating the navigator as an anchor, build it treating the navigator as a bias-discounted actor, and measure which predicts better on held-out outcomes. The differentiator is an engineering choice with a test attached, and the test is the arbiter.

One element comes closest to resisting the reduction, and I record it as the place to dig rather than as a claim. The egocentric-bias propagation of §4.5 — that error in the navigator's self-assessment is not independent of the navigator's model of how others perceive them, but contaminates it wherever the

navigator figures in the recursion — is a claim about correlated error structure across the belief state, a coupled prior a naïve I-POMDP user would not write down by default. I-POMDP's belief state is a joint distribution over the interactive state space and can in principle carry arbitrary correlation, so this too is expressible and is not a formal obstruction. But it is the one specification in the framework that departs from what the formalism's default use would produce, and if a formal separation result is ever to be sought, this is where it would be sought. Noted, not claimed; filed for future work.

2.4 Adjacent work

One recent system occupies a specific premise of this framework closely enough to require citation, though it is orthogonal to the headline. ASPECT (Shang et al., 2026) builds a behaviour-grounded communication profile of a single person from their data, and reports that its generated responses were preferred over self-report-based baselines in aggregate — with substantial variation across individuals — and that reviewing the profile led users to recalibrate their own self-assessments. This is an empirical instance of the navigator-bias premise that §4.5 develops — a behaviourally grounded model whose outputs are preferred to, and prompt correction of, that person's self-account — and, in the system correcting the user's self-model, an instance of model-correction running in the direction from system to user. It is a single-actor, single-layer, non-recursive system built for communication mimicry rather than relational prediction, so it does not approach the integrated architecture; but on the narrow point that a behaviourally grounded model can be preferred to self-report and can prompt its correction, it is corroborating prior work, and I cite it as such rather than as a competitor.

Finally, the phenomenon this framework proposes to support — that relational acuity materially shapes professional outcomes — is itself the subject of a mature behavioural-science literature that I draw on as evidence the phenomenon is real and structured, not as territory I contribute to. Work on social capital (Burt, 1992), on political skill in organisations (Ferris et al., 2005), and on interpersonal-perception accuracy (Funder, 1995) documents the performance consequences of relational modelling; theories of motivation — expectancy (Vroom, 1964), goal-setting (Locke & Latham, 2002), and self-determination (Ryan & Deci, 2000) — supply structured accounts of what actors want and why. RCL is not a contribution to that science. It is an engineering claim about what a system must represent to assist a user whose work is mediated by the responses of others, and it inherits from that literature the warrant that the thing being modelled has structure worth modelling.

2.5 The gap

The three programmes are complementary in their strengths, and the position this framework occupies is the cell none of them holds. Agent memory supplies persistence and continuous update but holds relationships as stored items in a single layer. Machine theory of mind supplies latent-motivation inference and recursion, but episodically, for an isolated agent, without a self-model, a structural field, or the historical record on which much of the inference depends. Integrated agent architectures supply the full integration of self, others, environment, and recursion — and, as §2.3 concedes, RCL is formally expressible as an instance of one of them — but that integration has been developed for an autonomous agent computing its own action, and never built for the open human domain, because the formalism requires a complete finite specification that such a domain cannot supply.

What RCL adds is therefore not a new representational primitive, and the contribution should not be read as one. It is the assembly, into a buildable system, of an other-model held as inferred wants, a served-human self-model held under dimension-specific distrust, and a structural field — conditioned on the accumulated record of an ongoing relationship and maintained over it, in support of a human navigator rather than in service of the system's own action. Two parts of that assembly carry the weight, and they are of different kinds. One is empirical and beyond the reduction's reach: that

conditioning want-inference on the accumulated record is necessary and large for prediction in this domain, because much of what makes behaviour interpretable is history-constituted (the keystone; Claim 4; §4.6). The other is an engineering achievement the reduction explains rather than undercuts: that the formalism's outputs can be obtained by bypassing the specification it requires, which is why a twenty-year-old integration becomes buildable now and not before (§1, §2.3).

I state the strength of this positioning carefully, because the field is converging on adjacent territory quickly and several near-neighbours have appeared within the last year. The claim rests on a non-systematic survey: on that survey I did not find a system occupying this cell, and I have named the nearest neighbours I did find rather than assert the cell is empty. A systematic review is companion work. Where the difference holds, it is one of kind rather than degree — not a better memory store, not a better theory-of-mind model, and not an autonomous social agent, but the assembly of these into a longitudinally maintained support system for an open human domain. Table 1 sets out the contrast across the dimensions that distinguish the programmes.

Table 1. RCL contrasted with the three adjacent programmes. The comparison is drawn on what each programme is built to do and what it conditions on, not on representational primitives: as §2.3 establishes, RCL is formally expressible within the integrated-architecture family, so the discriminating axes are purpose, conditioning regime, and specification demand rather than the kind of object represented. RCL cells carry the paper's own maturity tags; the other three columns are characterisations of published systems and are not so tagged.

Dimension	Agent memory (GraphRAG, Zep/Graphiti, Mem0, MemGPT)	Machine ToM (BToM, Rabinowitz; Strachan/Street)	Integrated agent architectures (I- POMDP, SOM, MetaMind, Wang et al.)	RCL (proposed)
What the output is for	Retrieval of stored information	Behaviour prediction, episodic	The agent's own optimal action	A prediction delivered to a human navigator — support, not action (<i>load-bearing</i>)
Conditioning regime	Stored items, timestamped; retrieved on query	Latent mental states inferred within the episode	Latent types / reward parameters, conditioned on the specified state	Want-estimates conditioned on the accumulated record; much meaning history-constituted (<i>working position; keystone</i>)
Specification requirement	Schema fixed; content populated by extraction	Model of rational action assumed; environment specified	Complete finite specification of states, types, dynamics required	Specification bypassed — approximate inference from unstructured evidence, no complete model constructed (<i>working position; Claim 4</i>)
Horizon	Continuous state (Zep), but as item-storage	Episodic / one-shot	Episode or planning horizon; multi-agent	Longitudinal, over the accumulated record of the relationship (<i>working position</i>)
Layers represented (<i>coverage, not novelty</i>)	One — partial relational, or contextual as corpus	Relational only (the other agent)	Self + other + environment + recursion, integrated	Navigator + relational + contextual, integrated (<i>working position</i>)
Role of the self	Not a distinct element	Not modelled —	Reward-bearing	First-person model of the

Dimension	Agent memory (GraphRAG, Zep/Graphiti, Mem0, MemGPT)	Machine ToM (BToM, Rabinowitz; Strachan/Street)	Integrated agent architectures (I- POMDP, SOM, MetaMind, Wang et al.)	RCL (proposed)
		neutral observer	decision-locus, or proxy anchor	served human, distrusted on evaluative dimensions (<i>distinctive parameterisation; settled by §6.1 ablation</i>)
Recursive belief	Not represented	Higher-order reported	Nested by construction	Bounded to 2–3 levels, on behavioural grounds (<i>working position; §5.4</i>)

3. The navigator and the tool

One commitment governs the architecture and must be stated before it: the system is not the navigator. The user is the navigator; the system is the instrument through which they navigate. This is a claim about the locus of agency — who acts in the world, with a stake in the outcome — and it has direct consequences for what the system represents and how. It should not be confused with a claim about the locus of trust: that the user holds agency does not make the user's self-account the schema's reliable reference point, and §4.5 argues it is the opposite. The user is the actor; the user is also, within the representation, one more actor to be modelled.

The user acts in a world of other people, human-constituted organisations, and non-human structural forces. The system augments that activity; it does not undertake it. The immediate consequence is epistemic: much of what the system holds about the actors in the user's world arrives filtered through the user. When the user records that a counterparty has grown distant, that judgement already carries the user's perception, position, and blind spots. The relational layer is therefore not a record of the social world but a record of one participant's view of it — and the participant supplying that view is not a transparent instrument but a modelled actor with their own wants and distortions, a point that becomes structural in §4.5.

This filtering is a property of the present, not a permanent feature. As always-on voice, meeting capture, and video become routine inputs, the system increasingly has observations of its own, and its independent read on an actor may diverge from the user's — sometimes usefully. What stays fixed under that change is the division of roles: the system sharpens the instrument through which the user perceives and acts; it is not a second agent loose in the user's social world. The consequence of that changing access matters below, because a system that only ever knew what the user told it could not, in principle, correct the user — it could only return the user's own model. A system accumulating independent observations can.

Why the system models recursively

The commitment also explains a design choice that would otherwise look like biomimicry, and it is worth being exact about the warrant, because the framework's claims are functional rather than cognitive. The terrain the user navigates is composed of other minds — actors who are themselves modelling the user, forming estimates of what the user wants, and adjusting as the user acts. To assist navigation of that terrain, the system must participate in the same logic: it must model actors who are modelling actors. A tool that did not represent recursion would be modelling the wrong object.

This is the point at which the role of cognitive science in the paper must be stated honestly, because the framework draws on established findings about human social cognition at several load-bearing points — the self–other knowledge asymmetry and egocentric bias in §4.5, cognitive-hierarchy depth in §5.4, reference-dependence in §5.5 — and it would be evasive to present these as mere motivation. They are not invoked to suggest the system thinks as a human thinks. They are relied upon as established findings about the humans the system models and serves: facts about the actors whose behaviour the relational layer predicts, and about the user whose self-reports the navigator layer ingests. An engineering system built to model humans must encode valid findings about those humans; importing them is therefore obligatory, not ornamental. The claim throughout is functional — this is what the system must represent to predict well — and at no point a claim that the system computes by the mechanism the finding describes. Where the distinction bears weight I mark it; here I state it once as the governing convention. The only point about human cognition the architecture treats as foundational, rather than as an input to be modelled, is the uncontested one that human social navigation proceeds by recursive mutual modelling — and that is a statement about the structure of the terrain, not about how either the human or the system traverses it.

3.1 The augmentation sequence

What the tool does for the navigator is not a single function but an ordered set, acquired as the quality of the system's model rises. I distinguish four developmental stages.

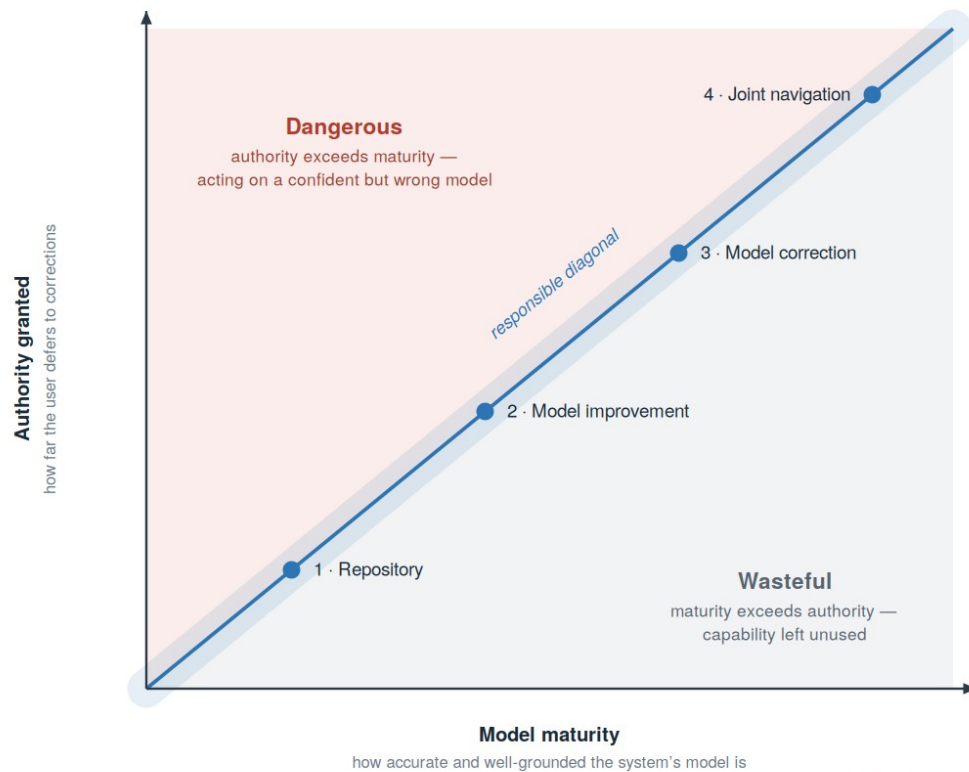
- **Repository.** The system holds more about more actors than the user can keep in mind, and returns it when relevant. This stage requires no independent model; its value is capacity and recall.
- **Model improvement.** The system begins to detect when the user's picture of an actor has gone stale, to flag what is missing, and to surface contradictions between the user's stored assumptions and incoming evidence.
- **Model correction.** With sufficient accumulated observation, the system's independent model of an actor may become more accurate than the user's, at which point the tool actively corrects the user's assumptions rather than merely extending them.
- **Joint navigation.** At the limit, the relationship becomes a shared undertaking: user and system each holding part of the model, neither simply deferring to the other.

A note on vocabulary, because it bears on a distinction the paper rests on elsewhere. The differentiator stated against the integrated agent architectures (§2) is that RCL offers *support rather than autonomous action* — the system predicts and the human decides; the system is never a second agent acting on its own account in the user's social world. The terminal stage above is therefore named *joint navigation*, not "collaboration," advisedly: even at its most capable, the relationship remains one of an instrument and the human wielding it, jointly holding a model, not two agents independently acting. The system's authority grows; its standing as a tool rather than an actor does not change. This holds across all four stages and is the through-line of the section.

The progression from holding the model, to improving it, to correcting it, is the framework's most consequential capability and its most ethically loaded one. A tool that corrects the user is valuable only insofar as its model is in fact more accurate, and that must be earned through a demonstrated track record rather than assumed from the system's confidence.

Two quantities move along this sequence, and the central design claim concerns their relationship: the maturity of the system's model (how accurate and well-grounded it is) and the authority the user grants it (how far the user defers to its corrections). The responsible trajectory is the one on which authority

tracks maturity — the user extends trust only as the system earns it. Figure 1 makes the two axes explicit and identifies the two off-diagonal failure regions; it is an illustrative schema rather than a derived result, and the placement of the four stages along the diagonal is indicative, not measured. Granting authority a model has not earned is the dangerous failure: acting on a confident but wrong model. Withholding authority a model has earned is the wasteful failure: capability left unused. The framework's ethical commitments, developed in §7, are largely commitments to remaining on the diagonal.



Schematic — illustrative, not derived

Figure 1. The maturity–authority plane. The four augmentation stages are waypoints on the responsible diagonal, where authority tracks model maturity; the off-diagonal regions are the two failure modes. The figure is illustrative, not derived: the diagonal and the stage placements indicate the intended relationship, not a measured one.

4. The three-layer architecture, maintained over a record

The architecture rests on the keystone stated in §1 (Claim 3), and this section is its specification. A system supporting response-mediated work predicts worse if it omits any of three elements — the user themselves (the *navigator layer*), the other actors in the user's world (the *relational layer*), and the structural conditions that bound what is achievable (the *contextual layer*) — or fails to let each condition the recursion, or severs the integrated model from the accumulated record of the relationship. The three layers answer *what must be represented*; the maintenance condition (§4.6) answers *what must be true of the representation over time*, and it is co-equal with the layers rather than an operational afterthought. The degradation predicted by the keystone should fall hardest in the situations where the missing element carries the most weight.

Behind the tripartition stands a stronger pair of claims, which I hold as an *organising hypothesis* rather than as the contribution. The first is that the three layers are jointly *exhaustive*: nothing that determines a navigation outcome falls outside them. The second is that they are mutually *irreducible*: no layer can be reconstructed from the others without predictive loss. These principles partition the architecture, and I give the irreducibility arguments as I specify each layer below. But the contribution rests on the longitudinal keystone (§1, §2.5), not on exhaustiveness or irreducibility, and the distinction governs how much weight a reader should place on each. Were the tripartition shown to be non-exhaustive — a determinant of outcomes found that fits no layer — the keystone would survive, augmented by whatever the missing element turned out to be. Were a layer shown to be ablatable without predictive loss, that is the keystone itself failing. The organising hypothesis is the more decorative of the two; the keystone bears the load. Both are tested in the ablation programme (§6) rather than asserted.

4.1 The relational layer

The relational layer holds models of the other actors in the user's world. Its defining commitment is what it models: not facts about actors but their wants. What predicts a counterparty's behaviour is less their job title or the record of past meetings than what they are trying to achieve — whether the distant supplier wants to be paid promptly or to be treated as a valued partner; whether the cautious budget-holder wants to save money or to avoid blame if a decision fails. The wager of the layer is that inferred motivation predicts behaviour more powerfully than the facts on record about an actor; I state that wager precisely, and qualify its strength, in §5.1. The formal apparatus for representing wants — how they are held, reweighted, and resolved when they conflict — is the subject of §5.

The layer's scope is not restricted to individual human actors. A navigator also models organisations, institutions, cultural norms, and abstract social forces — "head office," "the market," "what the regulator will tolerate." These entities have no single mind, yet they behave in ways that can be partially predicted from their structure, and a navigator reasons about them using inferential processes analogous to individual theory of mind. The relational layer accordingly admits any entity whose behaviour can be partially predicted from inferred, motivation-like structure, and models it with the same machinery.

4.2 The navigator layer

The navigator layer represents the user: their goals, constraints, and resources as these bear on the navigation problem. It is the element existing systems omit entirely, and its absence is more damaging than it appears. Whether an approach will succeed depends not only on the actors being navigated but on the navigator's own position — the credibility they hold with a given counterparty, the targets they are working toward, the resources that determine which moves are open to them. These are not background to the relational calculation; they are half of it.

The navigator layer carries two kinds of content, and the distinction between them matters for §4.5, though it is not the distinction one might expect to do the work. The first kind is the navigator's own motivational structure, which has the same form as that of any other actor: a persistent want-set, reweighted by situation, with wants at different horizons that can pull against one another. The self must be modelled with the same motivational apparatus applied to others. The second kind is the navigator's objective and perceived attributes — track record, network reach, resources, standing, and how others read their credibility — some of which bear on outcomes directly and some only through how others perceive them. The reliability of the user's own account of each kind is not uniform, and is not well predicted by which kind the content belongs to; how reliability actually partitions, and what the schema should do about it, is the subject of §4.5.

The navigator layer resists reduction to the relational layer, which supports the irreducibility hypothesis: a constraint on the navigator's resources forecloses options regardless of what any actor believes about the navigator. Some of its content acts on the world without passing through anyone's beliefs, and so cannot be reconstructed from a model of others' mental states, however complete.

4.3 The contextual layer

The contextual layer holds the structural and environmental conditions that bound what is achievable independently of what any actor wants. A valued supplier cannot be paid from cash that does not exist; a perfectly chosen champion cannot move a deal through a spending freeze. Budgets, governance structures, regulatory regimes, market conditions, macroeconomic shocks — these are not preferences to be modelled but limits on the field of play. A model that omits them produces advice that is socially astute and practically impossible.

The contextual layer resists reduction for a reason symmetric to the navigator layer's, and so supports the irreducibility hypothesis from the other side: its constraints hold regardless of belief. A relational model, however complete, cannot represent a fixed budget, because the budget is not a fact about any actor's mental state; it is a fact about the field. One boundary case complicates this cleanly drawn line, and I flag it as unresolved. Some structural constraints are objectively real (the budget); others are socially constructed (a norm about, say, the appropriate seniority for a given role) — structural in their force but constituted through collective belief. The latter sit at the boundary between the contextual and relational layers, structural in effect yet mediated through the very mental states the relational layer models. The treatment of socially constructed structural constraints is an open problem (§6.5).

4.4 Interdependence

The three layers are necessary, but they are not independent, and the dependencies carry predictive weight — they are the concrete content of the keystone's requirement that each layer condition the recursion. To represent the three layers' contents but not the dependencies between them would be to mispredict in exactly the situations where the layers pull on one another, which are the consequential ones. Four interaction effects are central.

First, the navigator's objective attributes partly constitute other actors' models of the navigator: a counterparty's assessment of the navigator's credibility is conditioned by facts about the navigator. Second, contextual conditions condition other actors' assessments: an actor weighs the same proposal differently when the organisation is thriving than when it is under stress. Third, contextual constraints bound the navigator's available actions directly, whatever any actor prefers. Fourth, the navigator's goals determine which contextual constraints are salient: which limits matter depends on what the navigator is trying to do.

The consequence for design is that the architecture cannot be built layer by layer in isolation. Figure 2 shows the arrangement: the navigator and the actors they navigate, both situated within the contextual conditions that contain them, modelling one another reciprocally, with the whole representation maintained over the accumulated record of the relationship.

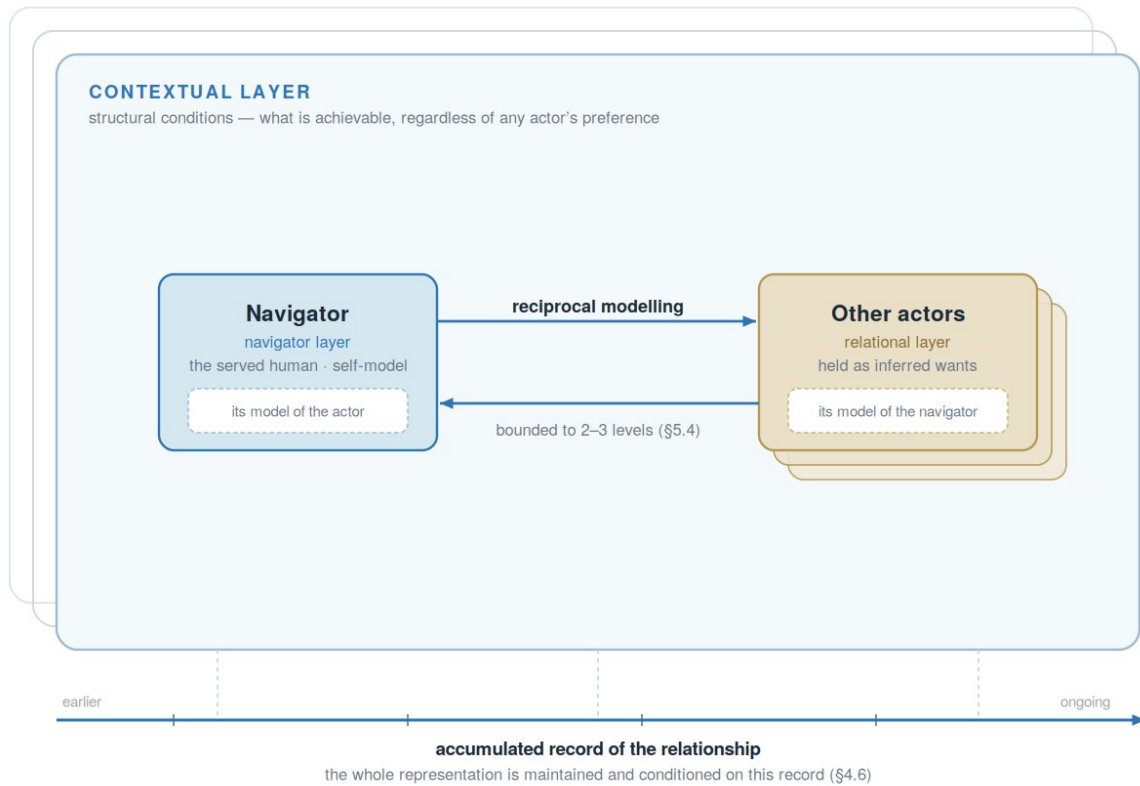


Figure 2. The three-layer architecture, maintained over a record. The navigator and the actors they navigate sit within the contextual layer that bounds them, and model one another reciprocally (bounded to two to three levels, §5.4); the whole representation is maintained over the accumulated record of the relationship.

4.5 How the layers relate to evidence

A structural property concerns not what the layers contain but how their contents are known, and it carries a result that is easy to get backwards. Existing memory systems model the user's own facts and history. This framework inverts that for the relational layer: it models other actors as seen through the user's observations — the *user-as-modeller inversion*. The data so obtained is subjective, incomplete, filtered through the user, and partially unverifiable, which is why relational-layer elements are held not as flat assertions but with confidence levels, with provenance distinguishing user-encoded claims from system-inferred ones, and with bias annotation.

The result that is easy to get backwards concerns the navigator layer. The inversion invites an assumption — that the first-person navigator layer is the schema's reliable reference point, since the user knows themselves best, and that uncertainty is concentrated in the third-person relational layer. This is the assumption the navigator-as-modelled-actor commitment (§2.3, §3) rejects, and the rejection has a specific shape that a blanket "the user is unreliable about themselves" would get wrong. Self-knowledge is not uniformly poor; it is poor on particular dimensions, and the boundary does not fall where §4.2's two content kinds fall.

The relevant finding is the self–other knowledge asymmetry (Vazire, 2010), which holds — about the humans the system models and serves, per the convention of §3, not about how the system computes — that a person is the *better* judge of their own low-observability, low-evaluative internal states (what they feel, intend, privately want) and the *worse* judge of their highly evaluative, ego-involved attributes (how competent, credible, or well-regarded they are), precisely where ego-protective distortion

operates and where others, observing from outside, see more clearly. The construct correspondence has to be argued rather than assumed, because SOKA was established on personality traits and the navigator layer holds wants and standing, not traits. The argument is that SOKA's two moderators — observability and evaluativeness — are general properties of self-knowledge rather than artefacts of the trait domain: a want that is private and not ego-loaded is reliably self-reported for the same reason a low-observability, low-evaluative trait is, namely privileged access without ego-protective pressure; and a want or attribute that is highly evaluative is unreliably self-reported for the same reason an evaluative trait is, namely that ego-protection distorts it. The correspondence runs through the moderators, which plausibly generalise, not through the trait construct, which does not transfer.

Drawn this way, the reliability boundary cuts *across* §4.2's content kinds rather than along them, and this is the point the naive version misses. Some of the navigator's internal wants are low-evaluative — and the user is a good source on these. But some internal wants are heavily ego-involved — to be admired, to win, not to be seen to fail — and these are *both* internal *and* evaluative, so being a "want" does not protect them: they are exactly the wants self-report distorts, and they are disproportionately the wants that drive high-stakes behaviour. Equally, the navigator's standing and credibility are evaluative attributes squarely in SOKA's territory, where the user is the worse judge. So the design rule is not "trust the navigator on wants, distrust on attributes"; it is "trust the navigator on low-evaluative internal states, and discount everything evaluatively loaded, whether want or attribute." And the trust is relative, not absolute — SOKA's reliability on internal states is itself noisy, so the rule is a heavier-or-lighter weighting, not a true-or-false switch.

One further class of navigator content is the most consequential and the least well served by SOKA directly: the navigator's beliefs about *how others perceive them* — meta-perceptions, the user's guess at a counterparty's reading of the user's credibility and position. SOKA concerns a person's accuracy about their own traits, not their accuracy at predicting others' views, so it does not directly license treating meta-perceptions as reliable or unreliable. A second established finding does the work here, and only to one hop: egocentric bias in adult perspective-taking (Keysar et al., 2003; Birch & Bloom, 2007), in which a person's own knowledge colours their model of another's mental state. These studies concern reasoning about what another believes of an external situation, not meta-perception as such; that the same egocentric intrusion reaches the navigator's estimate of how others see them is a generalisation I adopt, not a finding they establish directly. Its magnitude is in any case contested — the curse-of-knowledge effect has been argued small under replication, and is conditional on the perceiver having a plausible warrant to project their own knowledge — so I rely on its direction, not its size. This licenses exactly one claim — that the navigator's *self-supplied* meta-perceptions are egocentrically contaminated where they enter the schema. It does not license a claim about the system's recursion. Wherever the navigator figures as a modelled party deeper in the recursion, that contaminated input propagates by ordinary error propagation — an architectural fact about how an erroneous input travels through a model, not a further cognitive finding, and needing no citation. The distinction matters because the propagation is not fixed: it attenuates exactly as the system acquires independent observation of how others treat the navigator, observation that does not route through the navigator's self-report (§3). A system that only ever knew what the user told it would propagate the contamination undiminished; a system accumulating its own evidence dilutes it.

These meta-perceptions are not incidental content. What conditions a counterparty's response to the user is largely how that counterparty reads the user's credibility and standing — so the navigator inputs the recursion most depends on are disproportionately the evaluative, meta-perceptual ones the user reports least reliably. Precisely the navigator content the prediction most needs is the content the user is the worst source on. That is the sharp form of the navigator-discount: the navigator content the prediction most depends on is the content the user is the least reliable source on. Even granting that

the egocentric-bias effect is modest, its locus — falling on precisely these high-leverage meta-perceptual inputs — is what makes the discount a structural feature of the navigator layer rather than a refinement.

A final refinement keeps the unknown magnitude where it can be measured rather than baked into the architecture. The driver of self-model unreliability is not relational distance but affective involvement, for which distance is only a proxy — a proxy that holds for benign cases (one is more objective about a stranger than about oneself) and breaks for adversarial ones (one is not objective about an enemy, who may be distant yet a locus of intense affect). The self is simply the permanent high-affect endpoint of that scale. I therefore handle the navigator/relational difference not as a structural difference between two kinds of model but as a difference in the *weighting* of one shared mechanism — confidence, provenance, and bias annotation applied throughout, with a heavier bias weight on self-reported content and heaviest on self-reported evaluative content. Keeping the difference a weight rather than a structure leaves the magnitude — how steeply affective involvement degrades a model — as a calibratable parameter the validation programme can estimate, rather than a constant the architecture commits to and cannot correct if wrong.

A recent applied study corroborates the locus specifically, with a caveat that sharpens rather than weakens it. Shang et al. (2026), reviewing where a behaviourally grounded profile of a person's own communication diverged from that person's self-ratings, found the divergence surfacing an aspirational self-image — participants rating toward how they preferred to be seen, while the record showed the enacted, audience-shaped behaviour — and doing so on evaluatively loaded, self-presentational content rather than across the board: the evaluative locus this section isolates, observed in a workplace sample. The caveat is that the same study found the behavioural model carried its own distortion, systematically over-rating socially desirable traits — which is exactly why the design weights the self-account down on evaluative content rather than substituting a behavioural estimate for it. The corroboration is for a graded discount, not a replacement; and it is a single small sample ($n = 20$), carried as support, not weight.

None of these findings is the framework's contribution, and none is asserted as true of the system's computation; they are relied on as established facts about the humans the system models, and they motivate a design choice. The choice itself — trust the navigator on internal wants, discount on evaluative self-assessment, propagate the discount through the recursion where the navigator figures, attenuate it as independent observation accumulates — is settled not by the findings but by the navigator-discount ablation of §6.1, which is the arbiter (§2.3).

4.6 Maintenance over the record

The three layers specify what the architecture must represent. A further commitment, of equal standing, specifies what must be true of that representation over time: it must be maintained over the accumulated record of the relationship. There is an easy argument for this and a load-bearing one, and only the second makes maintenance part of the theory rather than the engineering around it.

The easy argument is staleness: a model of what an actor wants, left un-updated, decays as the actor and their situation move on, and a stale relational layer is worse than none because it is confidently wrong. That argument is correct but weak — a reviewer can fairly answer that every database needs updating and call maintenance mere hygiene. The argument that makes maintenance load-bearing is different. Much of what makes an actor's behaviour interpretable is *history-constituted*: its meaning is carried by the record of what came before and has no signature in the present moment. A curt, three-word reply is hostility from one colleague and unremarkable habit from another; the message itself discloses nothing that separates the two, and only the accumulated record of prior exchanges resolves

which it is. A system reasoning from a perfectly current snapshot — every present fact correct — still misreads that reply, because what it means is not present in the snapshot. So maintenance is not the upkeep of a fresh copy of facts that exist now; it is the holding of the only thing that makes present observations interpretable at all.

This is why the relational layer's want-estimates are conditioned on the record rather than on the latest evidence alone, and the distinction from existing memory work must be kept exact, so that elevating maintenance does not quietly re-claim persistence as the contribution. The bare persistence of facts is already supplied by agent-memory systems (§2.1); they store, timestamp, and reconcile. What they do not do is condition the *inference of what an actor wants* on the accumulated record — and that conditioning, not the persistence beneath it, is the load-bearing commitment. Maintenance is therefore co-equal with the three layers in the keystone, but it is not a fourth layer: the layers answer what must be represented, while maintenance is a property of the whole representation over time, a commitment of a different category.

The record over which the model is maintained is the *accumulated* record, and the word is exact: it is not merely what the system has observed since deployment. It includes history the user supplies about a relationship that long predates the tool — the backfilled past, encoded after the fact — because much of the constituting history of a working relationship is older than any system's access to it. This carries a limitation I name rather than elide, because it bounds the architecture honestly: the conditioning fails *asymmetrically*. Where the constituting history is recorded and accessible — whether observed by the system or backfilled by the user — the model can recover its interpretive force. Where it is off-record, never encoded by anyone, it stays dark, and no quantity of present observation reconstructs it. The ceiling is the *unrecorded* past, not the pre-deployment past; the latter can be supplied, the former cannot. The same gap constrains a human who joined the relationship late — which is the point: the limit is a property of the evidentiary record, not of the system's nature. But it is a real ceiling, and it is sharpest exactly at the start of a deployment into a relationship whose past was never written down.

The maintenance condition is tested, alongside the layers, by ablation: build the model on the full accumulated record, and again on a record truncated toward an episodic snapshot that keeps only the present state, and measure whether prediction degrades — most sharply, the keystone predicts, on exactly the actions whose sense is history-constituted (§6.1).

5. Operationalising the relational layer

The architecture claim invites an obvious objection: that "model what each actor wants" is a slogan, not a specification — anyone can say it, and saying it commits one to nothing testable. This section answers the objection by specifying the relational layer concretely enough to be falsified. The aim is not to present a validated model — the validation is deferred to §6 — but to demonstrate that the layer can be given a form precise enough that evidence could count against it. Where a choice in that form is itself a claim that could be wrong, I mark it.

5.1 The primacy of wants — a qualified wager

The relational layer's foundational commitment is to hold actors as estimates of what they want rather than as stored facts (§§2.1, 4.1). That commitment — model motivation, not attributes — separates the layer from the agent-memory systems and stands on its own. A distinct and stronger claim sits behind it, and the two must be separated because only the first is foundational: the claim that, among the things one might model about an actor, inferred motivation carries the greatest predictive weight. I state this second claim as an ordering — what an actor wants outweighs the actor's model of how to

obtain it (including their beliefs about what others want), which in turn outweighs a residual of disposition, mood, and circumstance — and I hold it as a wager, not a grounded result, because the established literature does not straightforwardly support it and in places cuts against it.

Two tensions must be stated plainly. First, the theory-of-mind tradition the relational layer otherwise draws on does not establish the primacy of wants and cannot be cited for it. That tradition's foundational and most-studied probe is the false-belief task (Wimmer & Perner, 1983), which foregrounds *belief*, not desire, as the central and difficult inference in predicting behaviour; if anything it locates the hard problem in belief attribution. Theory of mind establishes that behaviour is predicted by attributing mental states; it does not establish that desires outrank beliefs in doing so. Second, the ordering runs against the structure of expectancy theory (Vroom, 1964), which holds that motivated action is a product of how much an outcome is wanted (valence) and the actor's belief that action will attain it (expectancy and instrumentality). On that account wants are necessary but not sufficient, and they combine multiplicatively with attainment-beliefs rather than dominating them: an intensely wanted outcome believed unattainable motivates no action at all.

The framework's structure partly absorbs these tensions, which is why I retain the ordering as a wager rather than abandon it. The relational layer does not deny that beliefs matter: the second-ranked variable — the actor's model of how they obtain what they want, including their beliefs about what others want and will do — is the expectancy-instrumentality-and-belief component, and the recursive apparatus of §5.4 is where it is represented. The contestable commitment is therefore not whether beliefs are modelled (they are) but the ranking — that the want term carries more predictive weight than the belief-and-strategy term.

That ranking is most defensible in a particular situation class, and I restrict its strong form there rather than assert it generally — but the restriction must be drawn so that it sharpens the claim rather than immunising it, and there is a real hazard of the latter. Where the actors' menu of actions and the mapping from action to outcome are roughly common knowledge — as they tend to be within a settled professional setting, where competent participants share an understanding of how things get done — the expectancy and instrumentality terms vary little across actors. The hazard is that, where those belief-terms barely vary across actors, they cannot explain cross-actor variance in behaviour, so the want term wins almost by construction — which would make the restricted claim close to a tautology and the genuinely contestable question, whether wants dominate beliefs *when both vary*, exactly the case the restriction excludes. I therefore do not rest the wager on the low-variance regime. The claim I submit to test is the strong one: that inferred wants carry more predictive weight than the belief-and-strategy term *in the mixed regime, where both terms vary across actors* — the only regime in which the ordering is non-trivial and the regime in which expectancy theory most directly contests it. Two falsifiable boundaries follow. The situation class (shared versus contested action-outcome mapping) must be classified independently of, and prior to, the outcome it predicts — by inter-actor agreement on instrumentality beliefs, elicited separately from behaviour and coded blind — so that a setting cannot be relabelled "contested" because want-primacy failed in it; the operational classifier is specified in §6. And the ordering is tested where both terms genuinely vary, not where the belief-term is held near-constant by construction. If the want term fails to dominate in the mixed regime, the wager fails in the setting that matters; if it dominates even where the action-outcome mapping is contested, the situation-class restriction was unnecessary and the claim generalises further than I have claimed.

I do not rest the wager on early applied experience; the prototype observation of §6.4 is consistent with it but carries no evidential weight, for the reasons given there. The residual term, despite ranking last by aggregate weight, has internal structure that makes it disproportionately consequential, and I return to it in §5.5.

5.2 A persistent want-set, reweighted — and conflict resolved at evaluation time

An actor's wants are not a list from which a situation selects one item. They are a persistent set, held simultaneously, that a situation reweights rather than replaces. The cautious budget-holder wants to save money, to back a winner, and to avoid blame, at the same time; what a looming audit changes is not which wants exist but which becomes loudest. Time horizon — whether a want operates over the immediate interaction, the current period, or the actor's longer arc — is a property of each want, not a separate axis on which wants are indexed. Situational variation is carried by the reweighting of a single persistent set, for which the conditioning apparatus of §5.3 is the mechanism.

A consequence of holding wants as a set is that wants conflict, and a central design claim concerns how a conflict is represented. A conflict between two wants is a typed relation — it records the wants in tension and the axis of their tension — but it does not store a fixed resolution, a settled fact about which want wins. Resolution is computed at evaluation time, conditioned on the situation, because the same conflict reverses outcome under different conditions: the actor who lets irritation win against a junior will suppress it entirely before the board. Storing "irritation beats restraint" as a constant would mispredict precisely the cases where the situation flips the outcome. Tension is a relation to be resolved in context, not a constant to be looked up, and a model that stores resolutions rather than computing them is mis-specified by construction. Whether the computed-not-stored treatment of conflict has antecedents I leave to the systematic review; the commitment stands on its design rationale regardless.

5.3 How situations act: a bounded inventory and three functional forms

If situations reweight wants, the model must say what about a situation does the reweighting, and how. Both answers are constructed to be falsifiable.

The "what" is a bounded, pre-declared inventory of conditioning variables — a closed set, named in advance. For the worked professional case the core variables are the relationship class of the counterparty, the attributed motivation of the counterparty, the stakes configuration, and the trajectory or reference state, with several further candidates (relative power, observability, structural position in the interaction) on a watch-list. The boundedness is not a convenience; it is a falsifiability discipline, and this is the point of the construction. An inventory that could be extended after the fact — adding a new variable whenever a prediction failed — would explain any behaviour and therefore predict none. Fixing the set in advance is what makes the model capable of being wrong.

The watch-list must therefore be handled so that it does not quietly reintroduce the extensibility the discipline forbids, and this requires three sets to be kept distinct, not two. The *frozen inventory* — the core variables — is the only set admitted to a confirmatory test, fixed before the test is run. The *watch-list* is an explicitly exploratory set, analysed only in a labelled exploratory phase, and a watch-list variable may never be invoked to rescue a confirmatory prediction after the fact. Promotion of a watch-list variable into the frozen inventory is permitted, but it constitutes a *new* model that must earn its prediction on fresh held-out data, not inherit credit for the case it was added to fit. A behaviour that no frozen variable accounts for is therefore a *result* — a falsification of the inventory's completeness — not a trigger to extend the set mid-test. The protocol that enforces this freeze is specified in §6; here I record only that completeness is a claim that can fail under test, and is treated as a testing target rather than an assumption.

The "how" is a set of conditioning functions: declared statements of the direction in which each variable moves a want's salience, its reference point, or the resolution of a tension. These functions, not the variable list, carry the falsifiable content, and they take three distinct forms. A *gate* acts categorically — a variable engages or silences a want, with no graded middle. A *dose* acts monotonically — the effect

scales with the magnitude of the variable. A *regime* acts as a threshold change in the character of the response, not merely its size — beyond a threshold, the actor's response function changes in kind. The shape a given variable produces is itself a claim that can be falsified: a variable predicted to act as a gate but found to scale smoothly with magnitude has been mis-specified. Distinguishing a threshold (regime) from a steep but smooth nonlinearity (dose) is a finite-data model-comparison problem, and the procedure for drawing that distinction is part of the validation specification (§6). The regime shape in particular is not a free posit; it is grounded in the established finding that human response to risk is reference-dependent and changes character — not just magnitude — at the boundary between gain-framing and loss-framing (Kahneman & Tversky, 1979). How multiple conditioning variables combine when several act at once is governed by where each attaches in the path from stimulus to response — what I term the *attachment map* — rather than by a fixed precedence ordering among them; I treat the attachment map as a proposed mechanism and defer its detailed form and validation to companion work.

5.4 Recursive modelling and its bound

Actors do not act only on what they want; they act on what they believe others want, including what they believe the navigator wants. The relational layer must therefore represent recursive belief — the navigator's model of an actor's model of a third party. This recursion is real and predictively necessary, but it has a shallow reliable floor, and the bound must be stated in the right units, because two distinct findings bear on it and they are easy to conflate.

The quantity to be bounded is the *system's* nesting depth — how many levels of "models that" the representation carries. Two findings could in principle set it, and only one is the right instrument. The first is the account of intentionality orders (Kinderman, Dunbar & Bentall, 1998), which concerns the deepest nesting of mental-state attribution a person can in principle entertain — put at roughly the fourth-to-fifth order. This is a claim about *capacity*: the ceiling of competence. It is contested, and it is the wrong instrument here, for a reason that has become sharper since it was proposed. The bound governs a system whose modelling substrate is a large language model, and that substrate is not evidently capped at the human ceiling: models have been reported to match adult human performance overall on higher-order theory-of-mind tasks and to exceed it at the sixth order (Street et al., 2024; though Ullman, 2023, cautions that such performance is fragile under perturbation). If the substrate is not capped at the human capacity ceiling, that ceiling cannot be what sets the bound. (I note in passing that Dunbar is a co-author of the work that established the limit — Kinderman, Dunbar & Bentall, 1998 — and of the Street et al. work reporting it exceeded; the capacity figure and the evidence of its being overtaken trace to the same source.)

The second finding is the right anchor, and it differs from the first in construct and in direction. The cognitive-hierarchy model (Camerer, Ho & Chong, 2004) concerns not the ceiling of competence but the typical depth of iterated strategic reasoning people actually perform — found to centre on the order of one to two steps of best-response. This is a claim about *behaviour*, not capacity, and it describes the centre of what actors do, not the limit of what they could. It is the pertinent anchor because the system is predicting the modelled actors' behaviour: the relevant quantity is how deep those actors actually reason, since a model representing strategic sophistication the actors do not deploy adds noise, not signal. I do not lean on a third distinction sometimes drawn between these findings — that they differ in *unit*, intentionality orders versus best-response steps — because in the strategic setting the units are closely commensurable: a best-response step corresponds to a level of nested belief about the strategic situation. The disaggregation rests on construct (capacity versus behaviour) and direction (limit versus centre), which is enough to carry it; the unit claim is not needed and would not hold.

Two specifications make the bound exact. First, the counting convention: to predict an actor who reasons to a strategic depth of d steps, the system must represent that actor representing others, which is one level deeper than the actor's own reasoning — depth $d + 1$. The system's nesting depth and the modelled actor's strategic depth differ by one, and the bound is stated in the former. Second, the anchor is the consequential tail, not the population mean. Camerer's central figure is a population average, but the actors who matter disproportionately in high-stakes interactions sit in the tail of greater sophistication, reasoning to roughly two steps rather than the average of about one and a half. Bounding on the mean and then adding margin "for the tail" would be having it both ways; the bound is set on the tail directly. An actor reasoning to two strategic steps requires the system to nest to three. The bound is therefore two to three levels of system nesting, corresponding to a modelled-actor strategic depth of one to two steps — held as a heuristic grounded in typical behavioural depth, not as a claim about what the human or the machine can do, and beyond which signal-to-noise collapses.

5.5 Regime change and the structured residual

A model built only from the apparatus so far would be accurate most of the time and badly wrong occasionally — and the occasional failures would fall precisely on the high-consequence cases. The reason returns to the residual of §5.1. That residual is not uniform noise. It concentrates in a small number of factors that are dormant almost always and occasionally dominate — the factors that trigger a qualitative change in an actor's response, the regime shape of §5.3 operating at the level of the actor's whole evaluation.

The most important such factor is a shift in how the actor frames their situation. An actor protecting a gain behaves differently in kind from the same actor trying to avert a loss; the trigger of the shift is the reference point relative to which an outcome is framed, not the magnitude of the outcome (Kahneman & Tversky, 1979; the roughly twofold weighting of losses against equivalent gains was quantified later, in Tversky & Kahneman, 1992). Trajectory or reference state is therefore not one localised lever among others but a frame-setting variable that conditions the whole evaluation, acting largely upstream by lowering the threshold at which a situation is perceived as threatening. Two cautions attach to importing this finding, and I state them rather than paper over them. The finding is established on risky monetary choice; its extension to a social-strategic sense of "protecting a gain" versus "averting a loss" is a generalisation the framework adopts, not one the original result licenses directly. And the apparatus presupposes that the reference point can be identified — a known difficulty even in monetary settings and a sharper one in a career or organisational frame, where what counts as the actor's reference point is itself an inference. The framework's operability depends on locating the reference point, and that location is part of what the validation must establish, not assume.

This has a methodological consequence the framework takes seriously, and it cuts against the validation as much as the model. A model fitted to an actor's average behaviour will capture the common regime well and the rare one poorly, because the rare regime is by definition underrepresented in the behavioural record — so such a model is reliable in ordinary conditions and fails exactly when the stakes are highest. Accuracy on the mean and accuracy when it matters are different objectives. The framework treats regime change not as noise to be averaged away but as a distinct phenomenon to be detected — and detection *before* the shift manifests, rather than explanation after it, is the demanding form of the claim and the one on which the framework's highest-value capability depends. Retrospective identification of a reference-point shift is easy and proves nothing; prediction is the test. Whether the shift can be detected in advance, and whether a validation built on records in which rare regimes are sparse can test that detection at all, are open empirical questions confronted directly in §6, not capabilities the framework claims to have demonstrated.

6. Testability and the validation programme

A framework paper that defers its empirical work owes the reader two things in place of results: a precise account of what would falsify each principal claim, and a programme credible enough that another group could execute it. This section supplies both, and it is written to a stricter standard than the rest of the paper, because the reframing to an engineering reference architecture (§1) raises rather than lowers the evidential bar: an engineering architecture is judged by whether the built thing meets its constraints, so the validation specification is load-bearing for the whole position.

Two disciplines govern what follows. First, a single *primary confirmatory test* is named and fully specified (§6.1), and every other test is subordinate to it (§6.2). This is not presentational; it is the control for multiplicity. A framework that spawns a dozen ablations and reports any one that reaches significance will manufacture support, so the programme commits in advance to one primary endpoint, a ranked hierarchy of secondary endpoints, and family-wise correction across them. Second, the *tests themselves* carry maturity tags, not only the claims. "Tested by ablation" is a claim about a design, and until the design is specified to the point of execution it is a working position like any other. Where a test is specified to runnable grade I say so; where a capability is not yet testable with obtainable data I say that instead, and do not let the existence of a falsification scheme stand in for a falsification that could return a negative verdict.

A note on what the programme can and cannot establish, stated once. The ablation programme tests whether the framework's components *improve prediction* in the target domain. It does not, and cannot, establish architectural novelty over the interactive POMDP — §2.3 concedes there is none to establish, since the components are expressible within that formalism. The empirical question is the live one: whether the arrangement predicts better than the alternatives a deployer would otherwise reach for. That is what these tests are built to answer.

6.1 The primary confirmatory test: the navigator-discount ablation

The primary test is the navigator-discount ablation, chosen as primary for three reasons: it defends the one content commitment the §2.3 positioning rests on (the navigator as a bias-discounted actor rather than an anchor); it requires only behavioural data, not a solved-recursion metric or independent want ground-truth, making it the most obtainable test in the programme; and it is the minimum increment that would convert the framework from an argued position into one carrying a demonstrated effect. I specify it to the grade at which it could be pre-registered and run.

Corpus and unit. A longitudinal corpus of recorded professional interaction with a bounded set of actors, documented decisions, and observable outcomes over time — candidate sources are discussed and their limitations conceded in §6.3. The unit of analysis is a prediction occasion: a point in the record, at time T , at which an actor is about to respond to the navigator, and the response is recorded and later observable.

Prediction target. A pre-specified, externally codable feature of the actor's response — for example reply versus non-reply within a fixed window, escalation versus non-escalation, concession versus none — coded from the record by annotators blind to the model arm and to the model's prediction. The target is the actor's behaviour, not any latent belief, for the reason given in §6.3: behavioural targets avoid the circularity that latent-want targets invite.

The two arms. Both arms receive the identical three-layer model and the identical record up to T ; they differ only in how the navigator layer's self-reported content is treated. The *anchor* arm takes the navigator's self-assessment of standing, credibility, and how others perceive them at face value,

propagating it through the recursion. The *discounted* arm applies the §4.5 rule — trust on low-evaluative internal wants, discount on evaluative self-assessment and on self-supplied meta-perceptions, with the discount propagated where the navigator figures as a modelled party. The contrast isolates exactly the navigator-discount and nothing else.

Baseline. Beyond the two arms, a baseline that a deployer would otherwise use — a strong long-context language model given the same raw record without the navigator-layer structure — so that the test reports not only anchor-versus-discounted but both against the representation-free alternative.

Primary metric. A proper scoring rule on the calibrated predictive distribution over the coded outcome — log loss as primary, Brier score reported alongside — chosen because proper scores reward calibration and cannot be gamed by confident misprediction. Discrimination (AUROC) is a secondary metric, reported but not decisive.

Smallest effect size of interest. The decision rule requires a minimum effect, and the programme commits to the *procedure* for fixing it while declining to fabricate a number in advance of the corpus. The smallest effect size of interest (SESOI) is the smallest improvement in the primary metric that would change a deployment decision — the threshold below which the discount, even if real, would not be worth carrying — and it is fixed, with its justification, at pre-registration, before any outcome is seen. This is what gives the test a genuine negative verdict: an improvement of the discounted arm over the anchor arm that is statistically present but below the SESOI counts as the discount *not* carrying weight, not as a small success.

Statistical test and decision rule. A pre-specified test on the paired difference in primary-metric scores across prediction occasions, with confidence intervals. The decision rule is two-sided and committed in advance: the discount is supported only if the discounted arm beats the anchor arm by at least the SESOI with a confidence interval excluding the SESOI from below; it is rejected if the interval excludes an effect as large as the SESOI; and the result is inconclusive (underpowered) otherwise, in which case the honest report is that the corpus could not adjudicate, not that the discount failed. A power analysis fixing the prediction-occasion count needed to detect a SESOI-sized effect is part of the pre-registration; if the obtainable corpus cannot reach that count, the test is reported as underpowered by construction rather than run to a foregone null.

Regime stratification. Prediction occasions are stratified at coding time, blind to outcome, into common-regime and rare-regime (reference-point-shift) strata, for the reason developed in §6.2 and §6.3: the primary confirmatory claim is made on the common-regime stratum, where the corpus can supply enough occasions to power the test; the rare-regime stratum is reported separately and treated as exploratory, because it cannot be powered (§6.3). This stratification is the concrete form of the concession that the framework's highest-value capability is not the one its first validation can confirm.

6.2 The keystone and the secondary battery

Subordinate to the primary test, and reported under family-wise correction with a pre-specified hierarchy and gatekeeping (a secondary endpoint is interpreted as confirmatory only if the primary test and the endpoints above it in the hierarchy have succeeded; below the gate, results are exploratory), the programme tests the remaining principal claims. I order them by the weight they bear and state, for each, what would falsify it and whether falsification is fatal or merely narrowing.

The keystone (integration and maintenance). The keystone is tested by ablation against the same corpus: build the full model, then remove a layer's contribution or truncate the record, and measure degradation on the primary metric. Two honest qualifications govern the design. First, the keystone's conjuncts are *entangled*, and the programme does not pretend otherwise. One cannot ablate "the

navigator layer conditions the recursion" without removing navigator content, so a fully orthogonal factorial is impossible. The programme separates what separates: it crosses two factors that genuinely dissociate — whether a layer's content is *present as flat input* versus *conditioning the recursion*, and the *depth of record* supplied (full accumulated record versus truncation toward an episodic snapshot) — and it marks the degenerate cells (a layer cannot condition the recursion if its content is absent) as confounded, attributing no clean effect to them. What the design can cleanly establish is whether conditioning-on-the-record and conditioning-the-recursion each add predictive value over flat presence; what it cannot cleanly separate is named as confounded rather than reported as a result. Second, the maintenance ablation is where the keystone's distinctive prediction lives: truncating the record toward a present-state snapshot should degrade prediction *most sharply on the history-constituted stratum* — the actions whose meaning has no present signature (§4.6) — and the test reports the record-depth effect separately on that stratum. The keystone fails, in whole or in part, if a layer can be removed or the record truncated without degradation beyond the SESOI: an ablatale layer is a redundant one (irreducibility failing — fatal), and a snapshot that predicts as well as the maintained record shows the longitudinal condition carries no weight (the keystone's centre of gravity failing — fatal). Exhaustiveness is tested differently and is not fatal: it fails if residual prediction error traces to a determinant that fits no layer, in which case the keystone survives augmented by the missing element (§4). I note the cost the sceptic identified: a *large* missing determinant would leave "you need at least these three" intact while falsifying "these three are what you need," so exhaustiveness-failure narrows the claim by an amount proportional to the determinant's weight, rather than leaving it untouched.

The want-primacy wager. Tested by comparing a want-weighted representation against a belief-and-strategy-weighted one on held-out outcomes, in the mixed regime where both terms vary (§5.1). This requires discharging the obligation §5.1 placed on this section: an *outcome-blind classifier of the situation class*. Settings are classified as shared- versus contested-action-outcome-mapping by inter-actor agreement on instrumentality beliefs — elicited from participants or coded by domain experts, separately from behaviour, and fixed before outcomes are seen — so that no setting can be relabelled "contested" because want-primacy failed in it. The wager is supported if the want term dominates in the mixed regime; it fails there if it does not; and if it dominates even in the contested class, the situation-class restriction was unnecessary and the claim generalises (§5.1).

Relations as latent motivational states. Fails if a model representing actors as stored facts and typed edges predicts as well as one representing them as inferred, continuously updated wants, on the same longitudinal data — in which case the simpler fact-based representation existing systems already implement should be preferred. This is the test that the want-based representation earns its additional complexity.

The conditioning-function shapes. Each declared shape is a separate falsifiable claim, and discharging the §5.3 obligation requires a stated model-comparison procedure for distinguishing them in finite data. A variable's three candidate shapes — gate (categorical), dose (monotone-graded), regime (threshold change in kind) — are fitted as competing models and compared by out-of-sample predictive score with complexity penalised; a variable declared a gate but better fit by a smooth dose, or a regime boundary better fit by a steep but continuous gradient, is a mis-specification revealed by the data. The threshold-versus-steep-gradient distinction is the hardest and is treated as the limiting case of the procedure, not assumed away.

Inventory completeness. Fails if behaviour systematically requires a conditioning variable outside the frozen inventory — if prediction cannot be brought to adequacy without post-hoc additions. The freeze protocol that makes this a genuine test rather than a repairable shortfall is specified in §6.3.

The maturity–authority precondition. The normative claim that authority should track accuracy is not empirical, but its precondition is: that the system's independent model can, with accumulated evidence, become more accurate than the navigator's. This fails if the system's model does not overtake the navigator's on any measurable horizon — in which case the correction and joint-navigation stages (§3.1) are unreachable and the framework terminates at model improvement.

6.3 Corpus, protocol, and measurement

The most tractable methodology is hindcasting on a corpus with known outcomes: build the model from the record up to T , predict the state at $T+1$ from context available at T , and compare against what is known to have happened. Three methodological hazards attend it, and each is met with a committed design choice rather than a hope.

Circularity. If an actor's wants are inferred from the same behavioural record whose behaviour is then "predicted," the procedure is circular — fit wants to explain behaviour, then claim to predict the behaviour the wants were fit to. The programme avoids this in its primary study by *not requiring want ground-truth at all*: the prediction target is behaviour, and want-estimates are treated strictly as model internals, never scored against an independent want measurement. Whether the inferred wants correspond to actors' actual wants is a separate construct-validity question, deferred to a later study and not entangled with the behavioural test.

The want-construct-validity study and its target-mismatch. When that later study is run, it must avoid a trap §4.5 sets for it. The naive design elicits the target actor's wants by questionnaire and treats them as ground truth — but the relational layer does not hold the actor's wants; it holds the *navigator's filtered estimate* of those wants, so the questionnaire would validate a different quantity than the architecture computes. The study therefore elicits the navigator-filtered estimate where that is the modelled quantity, restricts any self-report ground-truth to the low-evaluative internal wants on which self-report is reliable per §4.5 (excluding the evaluative wants the same section flags as distorted), and carries the residual unreliability of even that self-report as an explicit measurement-error term rather than treating it as clean truth.

The inventory freeze protocol. Discharging the §5.3 obligation: the conditioning inventory is partitioned into three sets before any confirmatory test. The *frozen inventory* — the core variables — is fixed and is the only set admitted to confirmatory analysis. The *watch-list* is exploratory, analysed only in a labelled exploratory phase, and may never rescue a confirmatory prediction after the fact. Any promotion from watch-list to frozen inventory creates a *new* model that must earn its prediction on fresh held-out data, not inherit credit for the case that motivated the promotion. A behaviour no frozen variable accounts for is recorded as a completeness failure, not used to extend the set mid-test.

The behavioural-testability boundary. One distinction is harder than the rest and I locate it honestly rather than overstate its reach. A non-response can arise by two mechanisms — the situation never engages the want (upstream suppression, no internal conflict) or the want is engaged but its expression is restrained (a felt-but-not-enacted gap) — and at the level of a single stimulus-action record these are identical. Separating them may be partially possible from channel-separated behavioural data where a reliable verbal tell dissociates the two, but it is conditional on the channels being separable, observed at sufficient granularity, and stable enough to have been learned, and whether those conditions hold is itself empirical. The programme treats the suppression/attenuation discrimination as testable only with internal-signal or sufficiently channel-separated data, defers it accordingly, and does not rest any primary claim on it.

Feasibility, stated not assumed. The credibility of a deferred-evidence programme rests on a corpus existing that meets the conjunction of requirements: longitudinal, multi-party, bounded actor set, documented outcomes, and — for the later want-study — independent want measurement, with enough common-regime occasions to power the primary test. Corporate governance records meet several of these (long tenure, documented decisions, bounded networks, observable outcomes) but supply candid want poorly and rare-regime occasions sparsely; a purpose-built email-plus-questionnaire corpus can supply elicited want but is harder to extend over years. I do not claim a single corpus is in hand that meets the full conjunction; I specify the requirements, identify the candidates and their shortfalls, and scope the confirmatory claims to what an obtainable corpus can actually support — which is the primary navigator-discount test on common-regime behavioural outcomes, with the rest reported at whatever power the corpus permits.

The rare-event limit. The framework's highest-value capability — advance detection of regime change (§5.5) — is the one its validation can least support, and the programme says so plainly rather than burying it. Rare regimes are by definition sparse in any behavioural record, so an aggregate test is underpowered exactly where the theory locates the value, and a detector that never predicts a shift scores well by base rates alone. The programme therefore (i) stakes its primary confirmatory claim on common-regime prediction; (ii) reports rare-regime strata separately, with precision-recall and lead-time metrics under class imbalance, as exploratory and underpowered; and (iii) treats advance regime-detection — together with the reference-point identification it presupposes, the §5.5 obligation — as a research target the framework does not claim to have demonstrated. Reference-point identification is itself unsolved: locating the reference point relative to which an actor frames a situation is an inference the validation must establish a method for, not assume, and until it does, regime detection cannot be tested even where occasions exist.

6.4 Proof of concept: ReactAI

I have a commercial interest in ReactAI, a prototype discussed here for provenance only; nothing in this section is offered as evidence, and no inference about the architecture's validity is drawn from it.

ReactAI is a stakeholder-prediction tool built on a deliberately incomplete and unstructured implementation of the architecture, used to model high-stakes interactions before they occur. In building and using it I observed behaviour consistent with the framework's central bet — that supplying a small number of well-chosen, honest facts about an actor, above all a candid read on what that actor wants, appeared to improve prediction, and that prediction quality appeared to decay as supplied facts went stale. The observation has no evidential weight, for reasons that remove it entirely: the implementation touched each layer only in a weak, unconditioned form; the inputs that produced the better predictions were well-chosen and honest by my own selection, so any improvement is as readily attributed to input quality and to my own expectancy as to anything architectural; and no controlled evaluation was conducted, nothing logged, nothing quantified. The episode is reported as the origin of the framework's central bet and the kind of behaviour that motivated it — nothing more. Whether the bet holds belongs entirely to the programme of §6.1–§6.3, and the staleness the observation suggested is a hypothesis for that programme to test, not a finding the prototype delivers.

6.5 Open problems

Several problems are unresolved and named as such. The exhaustiveness and irreducibility of the tripartition are the organising hypothesis, awaiting the construction-and-ablation tests above (§4, §6.2). The boundary between the contextual and relational layers — the status of socially constructed structural constraints, structural in force yet constituted through belief — is unresolved (§4.3). Advance detection of regime change, and the reference-point identification it presupposes, is the framework's

highest-value capability and the one its present validation can least support (§5.5, §6.3). The unrestricted form of the want-primacy wager remains open beyond the mixed regime to which §5.1 confines the staked claim. The suppression/attenuation discrimination is testable only with data the primary programme does not require (§6.3). And the longitudinal conditioning of the relational layer carries a structural ceiling the framework does not overcome: it recovers history that is recorded — whether observed or backfilled — but never history that is off-record, a limit sharpest at the start of a deployment into a relationship whose past was never written down (§4.6).

7. Ethical considerations

A system built to model people on a navigator's behalf carries ethical weight in proportion to what it can do, and I treat that weight here as a set of design constraints and responsibilities rather than as a separate moral essay. None of what follows is an argument against the framework; each is a commitment that accompanies it. One of the five reduces cleanly to a representational requirement the architecture can be held to; the other four I state as constraints and, where the framework cannot yet discharge them, name as open.

7.1 Correction validity

The highest-value end state — the system telling a navigator their read on someone is wrong — is also the most ethically loaded, and it is the one place where an ethical demand reduces cleanly to something the architecture must represent. A correction is warranted only insofar as the system's own model is more accurate than the navigator's, and that superiority has to be earned through an accumulated track record, not assumed because the system asserts it. The representational requirement that follows is concrete: the system must hold confidence in its own estimate *independently* of the navigator's confidence, must carry the provenance of each element — distinguishing what the navigator encoded from what the system inferred (§4.5) — and must surface a divergence as a flagged discrepancy for the navigator to adjudicate, never as a silent substitution. The governing picture is a plane on which two quantities move as the system matures: how accurate its model is, and how much authority the navigator grants it. The responsible path is the diagonal, where authority tracks accuracy. The two off-diagonal regions are the failure modes, and they are not symmetric: withholding authority from a system that has earned it merely wastes capability, whereas granting authority a system has not earned — acting on a confident model that is wrong — is the one that does harm. Keeping to the diagonal is the framework's central ethical commitment, and because it is a representational discipline it is auditable rather than aspirational. (*Working position; design constraint.*)

7.2 Epistemic autonomy

A navigator who comes to defer routinely to the system's corrections may lose acuity in the very faculty the system augments: sustained reliance on an augmentation can atrophy the capacity it supports. This is the characteristic risk of the correction end state, and I do not have a resolution for it within the present framework — the design choices that would mitigate it (calibrating how often and how forcefully the system corrects, keeping the navigator's own reasoning in the loop rather than supplanting it) are real but untested. I flag it as an open problem rather than claim it is managed. (*Open.*)

7.3 Third-party consent

The relational layer models people who have not consented to being modelled, and cannot readily be asked. Two features of the architecture bound the concern without dissolving it. First, the system models third parties only as seen through one navigator's observations: the estimates are navigator-filtered, partial, and held with explicit confidence and bias annotation (§4.5), not as ground-truth dossiers. Second, the goal is prediction, not simulation — the system is built to forecast what an actor will do, a falsifiable and correctable objective, not to impersonate them, which would be a more invasive artefact of a different kind. Neither feature confers consent, and I do not claim the concern is discharged; the design is one that keeps the modelled party's representation provisional and outcome-checkable rather than authoritative. (*Open; bounded by the provenance and prediction-not-simulation commitments of §3–§4.*)

7.4 Distributive and reflexive effects

An effective relational augmentation confers a material advantage wherever relational acuity determines professional outcomes, and that advantage has two social consequences the framework should own. For as long as access is unequally held — as it will be, at least initially — the capability is distributive: it advantages those who have it over those who do not, in settings where the difference is consequential. Once it is widely held, it becomes reflexive: the strategic field itself changes as other parties begin to model the fact that a navigator may be augmented, so the environment adapts to the tool's presence. Neither consequence is an argument against building the system, but both are responsibilities that accompany its deployment, and both warrant formal treatment beyond what I give them here. (*Open; flagged for fuller treatment.*)

7.5 Domain of application

The framework is scoped to *professional* response-mediated work, and the scoping is an ethical boundary rather than an incidental choice of domain. The same machinery — a maintained model of what an actor wants, conditioned on an accumulated record, used to anticipate their response — pointed at one's intimates rather than one's professional counterparts is a manipulation aid, and the asymmetry is sharp enough to draw the line at the domain rather than leave it to the user's discretion. The relationships the framework is built to support are ones in which the parties already model one another strategically as a matter of role — negotiation, management, advocacy, organisational navigation — and where anticipating a counterpart's response is a legitimate and expected part of the work. That is the terrain §1 restricts the framework to, and the restriction is load-bearing: the manipulation-of-intimates case is not a peripheral misuse to be discouraged but the reason the professional scoping is written into the claim from the outset. (*Design constraint.*)

8. Conclusion

Systems built on large language models begin each interaction stateless with respect to the world the work is conducted in: the people the user must act through, the user's own position among them, and the structural conditions that bound what is achievable. I have argued that this caps relational assistance at what a capable stranger could offer, that the cap is architectural rather than a deficit scale will close, and that lifting it calls for a specific arrangement of parts — a first-person model of the served navigator, other actors held as context-conditioned estimates of what they want, and the structural field that bounds the possibility space, all maintained over the accumulated record of the relationship. That arrangement is the Relational Context Layer, offered as a reference architecture in the engineering sense — components, layers, and a maintenance regime — not as a new representational grammar.

RCL reduces to a finitely nested interactive POMDP, and I have stated that mapping rather than resisted it. The formalism is two decades old; what was missing was not the representation but the means to solve and populate it from the unstructured social evidence of real work, and that is what large language models newly supply — which is why adjacent systems are appearing now, arriving at the feasibility rather than the idea. The contribution therefore sits where the reduction cannot reach, and two parts of it have independent life. The first is a specification bypass: because the output is a prediction delivered to a human rather than an action selected by an agent, the system asks for far less than I-POMDP deployment — no policy optimisation, recursion bounded at two to three levels, approximate inference in place of exact belief update — and so becomes buildable where the autonomous case was not. The second, which I stake the paper on, lies outside any representation: that much of what makes an actor's behaviour interpretable is history-constituted, carried by the record and absent from the present moment, so a system that severs want-inference from that record predicts worse — most sharply on the actions whose meaning lives in the history. That is the keystone, and I hold it as a working position to be settled by ablation, not as a result already in hand.

Two further commitments deserve restating at their proper strength. The navigator-discount parameterisation — trusting the served human's account of their own wants while distrusting it on the evaluative, ego-involved dimensions where self-assessment is least reliable, and carrying that discount through the recursion — is a parameter assignment no integrated-agent practitioner makes by default, and I treat it as a testable commitment to be decided by ablation (§6.1), not asserted. The delegation conjecture I hold more tentatively still, as a conjecture and not a finding: that the binding constraint on how much professional work is actually delegated to an AI system is not the capability of the underlying model but the absence of a maintained relational context and the cost of supplying it afresh each time. It is falsifiable in the same currency as the rest — hold model capability fixed, vary whether the context is maintained, and measure whether the share of real work a user will delegate rises with it; if it does not, the limit on AI's reach in this domain lies elsewhere than where I have placed it.

What I have tried to keep this proposal from being is a plausible story that cannot be wrong. The relational layer is operationalised to the point where evidence can count against it; the validation programme fixes corpus, baselines, metrics, effect-size thresholds, and decision rules in advance; and the navigator-discount ablation is named as the minimum increment that would move the keystone toward grounded or away from it. Several questions remain open and are marked as such: whether the three layers are jointly exhaustive and mutually irreducible, where the contextual and relational layers properly divide, and whether regime change can be anticipated rather than only recognised after the fact — the capability the present validation can least support. The prototype that prompted the framework carries no evidential weight and is reported only as provenance. The claim I am left making is narrow and, by intent, checkable: that supplying and maintaining the relational context of response-mediated work is a distinct and buildable thing, that it is the part the I-POMDP reduction does not absorb, and that whether it improves prediction is a question with a definite answer — one the programme of §6 is built to obtain.

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